

# Two-Sample Proportion Test in R

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## 1. Introduction and Objective

Suppose we have two independent populations with success proportions  $p_1$  and  $p_2$ . We want to test whether there is a significant difference between the two proportions.

$$H_0 : p_1 = p_2 \quad \text{vs.} \quad H_1 : p_1 \neq p_2$$

This is a two-sided hypothesis test for the equality of two proportions.

## 2. Sample Proportions

For each sample, the estimated sample proportions are:

$$\hat{p}_1 = \frac{x_1}{n_1}, \quad \hat{p}_2 = \frac{x_2}{n_2}$$

where  $x_i$  is the number of successes and  $n_i$  is the sample size.

## 3. Pooled Proportion under $H_0$

Under the null hypothesis  $H_0 : p_1 = p_2$ , the common (pooled) proportion is estimated by:

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

## 4. Standard Error (SE)

The standard error of the difference between two proportions under  $H_0$  is:

$$SE = \sqrt{\hat{p}(1 - \hat{p}) \left( \frac{1}{n_1} + \frac{1}{n_2} \right)}$$

## 5. Test Statistic

The test statistic for testing  $H_0$  is:

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{SE}$$

Under  $H_0$ , this statistic follows the standard normal distribution:

$$Z \sim N(0, 1)$$

## 6. p-value

For a two-sided test:

$$p\text{-value} = 2 \times [1 - \Phi(|Z|)]$$

where  $\Phi(\cdot)$  is the cumulative distribution function (CDF) of the standard normal distribution.

## 7. Confidence Interval for the Difference

A  $100(1 - \alpha)\%$  confidence interval for the difference  $(p_1 - p_2)$  is given by:

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \times SE$$

## 8. Decision Rule

Reject  $H_0$  if  $p\text{-value} < \alpha$ ; otherwise, fail to reject  $H_0$ .

## 9. Numerical Example in R

In this example, we have:

$$x_1 = 45, \quad n_1 = 100, \quad x_2 = 55, \quad n_2 = 120$$

Listing 1: Detailed two-sample proportion test in R

```
1 #####
2 # Two-sample proportion test in detail
3 # Example: prop.test(c(45, 55), c(100, 120), alternative = "two.sided",
4 #   correct = FALSE)
5 #####
6 # 1. Input data
7 x1 <- 45; n1 <- 100
8 x2 <- 55; n2 <- 120
9
10 # 2. Sample proportions
11 p1_hat <- x1 / n1
12 p2_hat <- x2 / n2
13
14 cat("Sample proportions:\n")
15 cat("p1_hat =", round(p1_hat, 4), " p2_hat =", round(p2_hat, 4), "\n\n")
16
17 # 3. Null and alternative hypotheses
18 cat("H0: p1 = p2 vs. H1: p1 != p2 (two-sided test)\n\n")
19
20 # 4. Pooled proportion (under H0)
21 p_hat <- (x1 + x2) / (n1 + n2)
```

```

22 cat("Pooled proportion (under H0): p_hat =", round(p_hat, 4), "\n\n")
23
24 # 5. Standard error under H0
25 SE <- sqrt(p_hat * (1 - p_hat) * (1/n1 + 1/n2))
26 cat("Standard error (SE) =", round(SE, 5), "\n\n")
27
28 # 6. Test statistic (Z)
29 Z <- (p1_hat - p2_hat) / SE
30 cat("Test statistic (Z) =", round(Z, 3), "\n\n")
31
32 # 7. p-value
33 p_value <- 2 * (1 - pnorm(abs(Z))) # two-tailed p-value
34 cat("p-value =", round(p_value, 4), "\n\n")
35
36 # 8. 95% Confidence interval for (p1 - p2)
37 alpha <- 0.05
38 z_alpha2 <- qnorm(1 - alpha/2)
39 CI_lower <- (p1_hat - p2_hat) - z_alpha2 * SE
40 CI_upper <- (p1_hat - p2_hat) + z_alpha2 * SE
41 cat("95% CI for (p1 - p2): (", round(CI_lower, 4), ",", round(CI_
    upper, 4), ")\n\n")
42
43 # 9. Decision
44 if (p_value < alpha) {
45   cat("Result: Reject H0 at alpha =", alpha, "\n\n")
46 } else {
47   cat("Result: Fail to reject H0 (no significant difference)\n\n")
48 }
49
50 # 10. Compare with R's built-in test
51 cat("-----\n")
52 cat("Comparison with R's prop.test() output:\n\n")
53
54 result <- prop.test(c(x1, x2), c(n1, n2),
55                     alternative = "two.sided",
56                     correct = FALSE)
57 print(result)
58
59 # 11. Additional explanation
60 cat("\nExplanation:\n")
61 cat("- X-squared value in prop.test() equals Z^2.\n")
62 cat("- df = 1 because we compare two proportions.\n")
63 cat("- p-value close to 0.902 indicates no significant difference.\n"
64 )
65 cat("- Confidence interval includes zero, confirming equality of
66   proportions.\n")
67 cat("- 'correct = FALSE' disables Yates continuity correction.\n")
68 cat("-----\n")

```

## 10. Interpretation of Results

- The computed p-value ( $\approx 0.902$ ) is very large, indicating no significant difference between the two proportions.
- The 95% confidence interval includes 0, confirming that  $p_1$  and  $p_2$  are statistically equal.
- The X-squared statistic reported by `prop.test()` is equal to  $Z^2$ .
- Setting `correct = FALSE` disables Yates' continuity correction, giving a closer match to the theoretical Z-test.